UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

Class 001 · Introduction to second and higher order DE with constant coefficients · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·12	Unit 2 deck 001, slide 12 (rendered .pptx)
T001 00:17:38	Unit 2 AV-summary transcript 001 (“Introduction to higher order differential equations, complementary function and particular integral”), segment start
ESA Dec-23 3a	Past paper: UE22MA141A Dec 2023, question 3.a
ESA Jan-24 s3 3d	Past paper: UE23MA141A Jan 2024 set 3, question 3.d

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5. The forms of a linear DE in the deck
6. Complementary function against particular integral
7. Every Unit 2 ESA question is CF + PI

1 Linear differential equations and their order

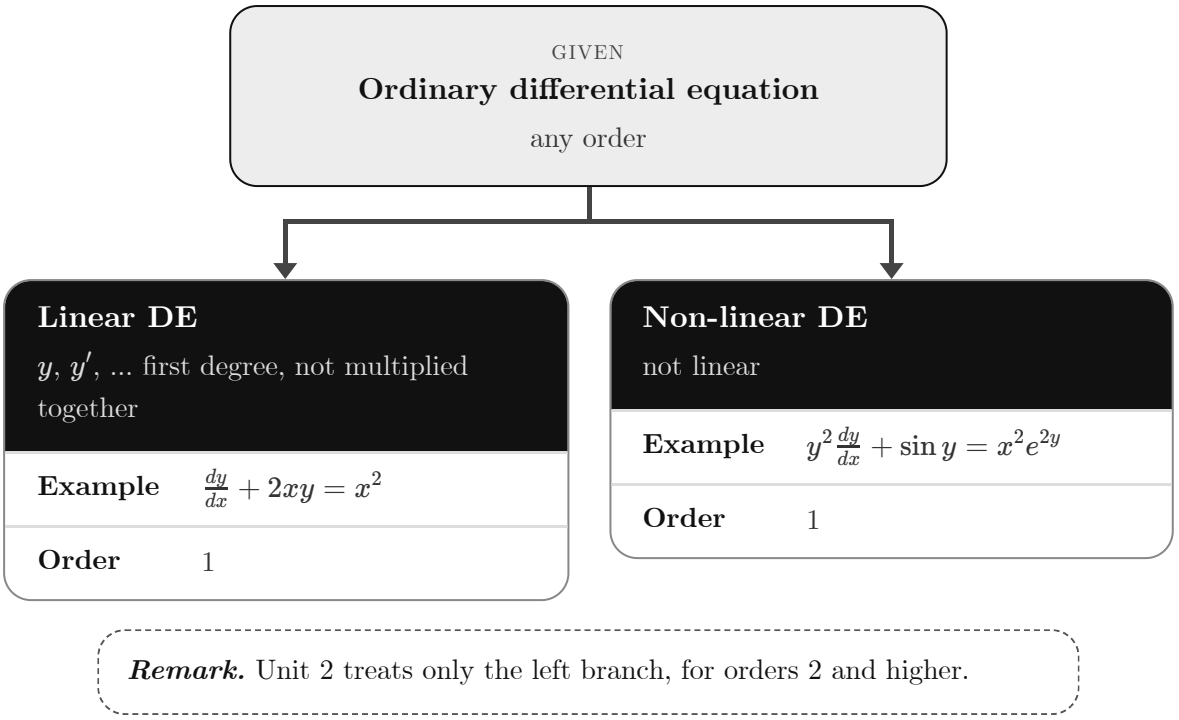
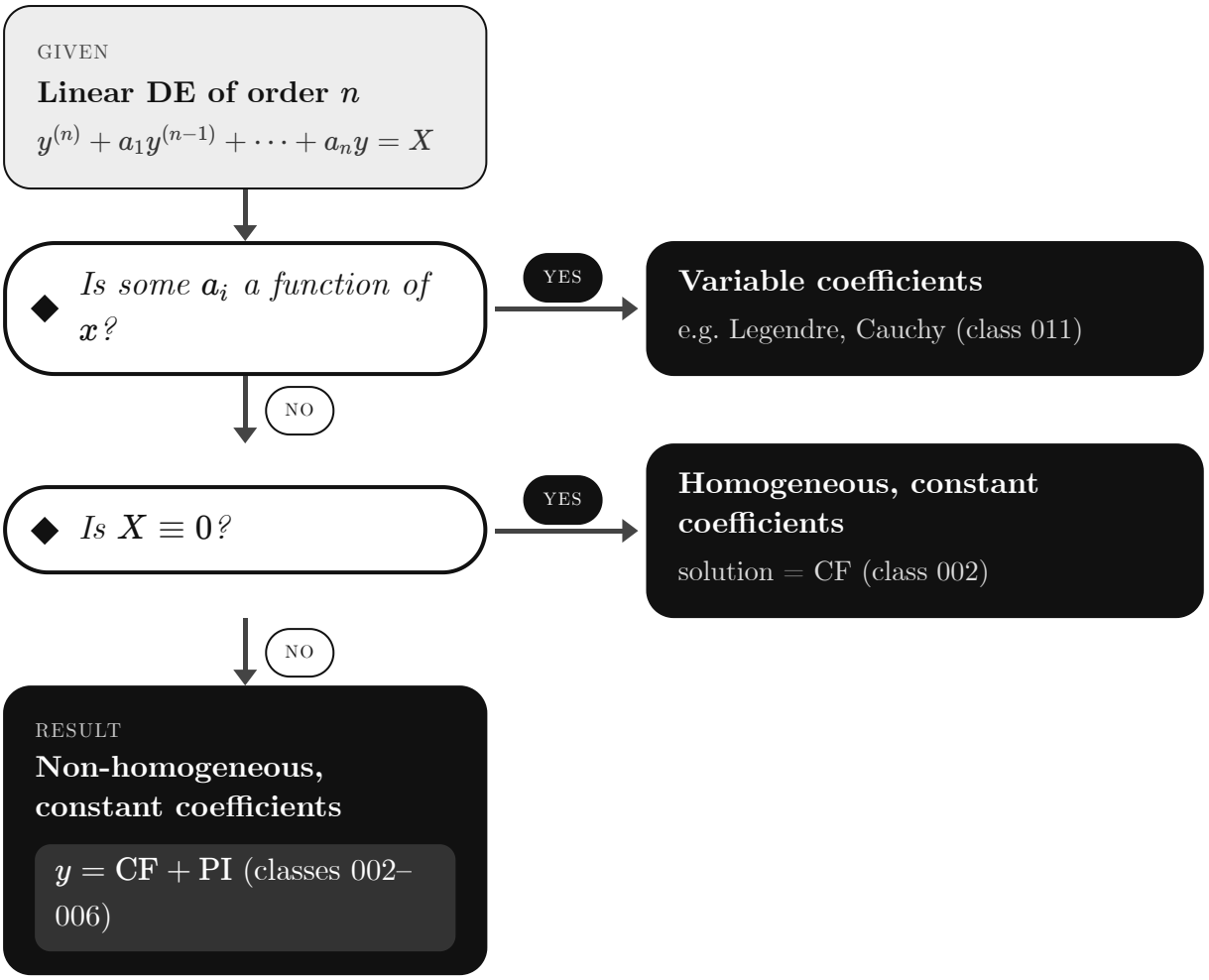


Figure F1.1. Classifying an ordinary differential equation (slide 001·5) ESA ★

S001·5 · T001 00:02:54 · T001 00:03:46

2 Constant or variable coefficients; homogeneous or not



Remark. Slide 001·10 draws the homogeneous / non-homogeneous split as a tree under "LDE with constant coefficients"; the example pair is $2y'' - 3y' + 9y = 0$ and $2y'' - 3y' + 9y = x \sinh x$.

Figure F2.1. Classifying a linear DE (slides 001·8–001·10) ESA ★★

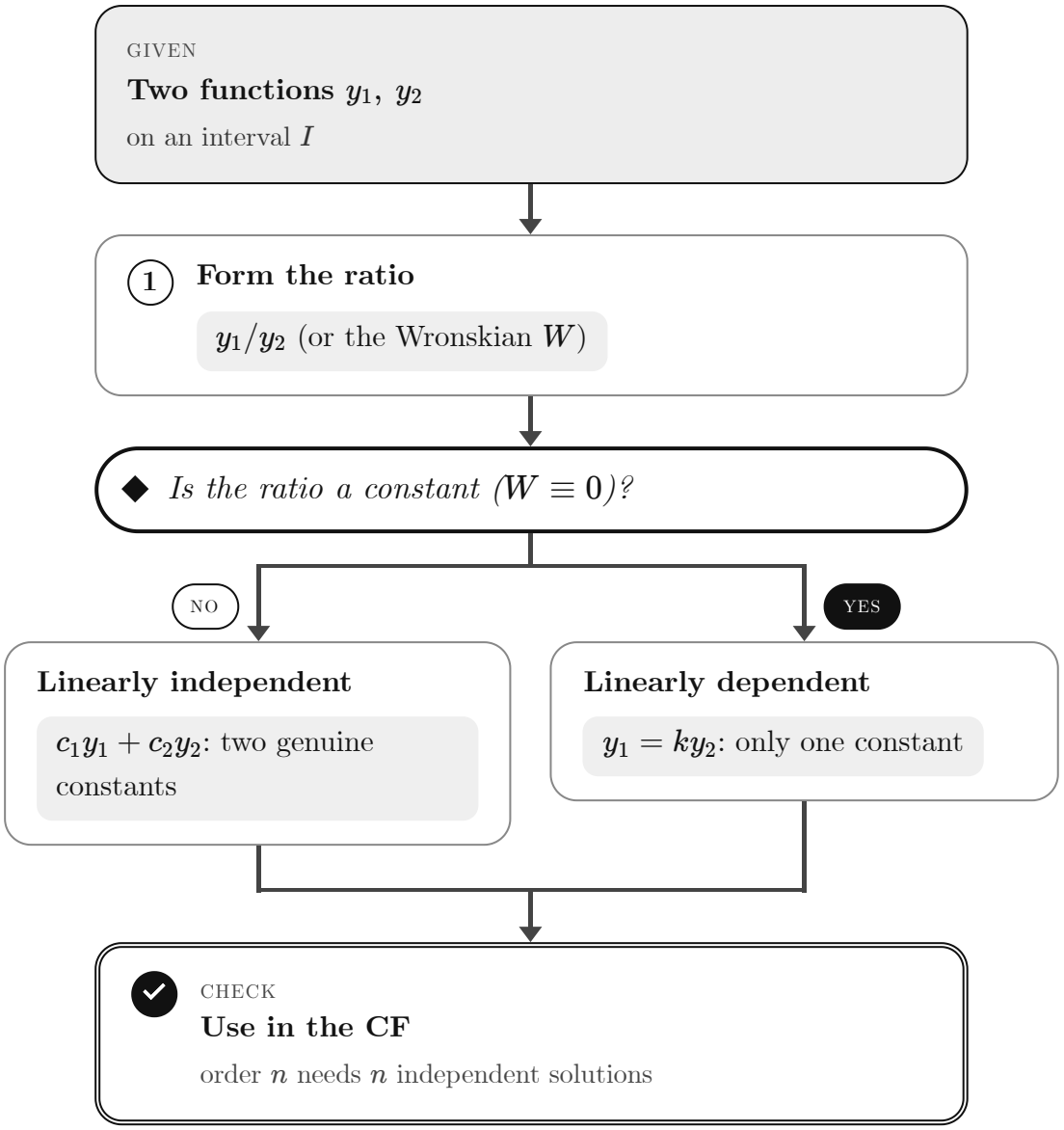
S001·8 · S001·9 · S001·10 · T001 00:10:11 · T001 00:13:13

The forms of a linear DE in the deck

Form	Coefficients	RHS X	Deck example	Order
First-order linear	P function of x	$Q(x)$	$\frac{dy}{dx} + 2xy = x^2$	1
Constant coefficients	numbers	x^2	$2y''' - 5y'' + 7y' + y = x^2$	3
Variable coefficients	functions of x	$x^2 \sin x$	$2x^3y''' - (5x^2+2)y'' + xy' + y = x^2 \sin x$	3
Homogeneous (const.)	numbers	0	$2y'' - 3y' + 9y = 0$	2
Non-homogeneous (const.)	numbers	$x \sinh x$	$2y'' - 3y' + 9y = x \sinh x$	2

S001·7 · S001·8 · S001·9 · S001·10

4 Linear independence and the complementary function

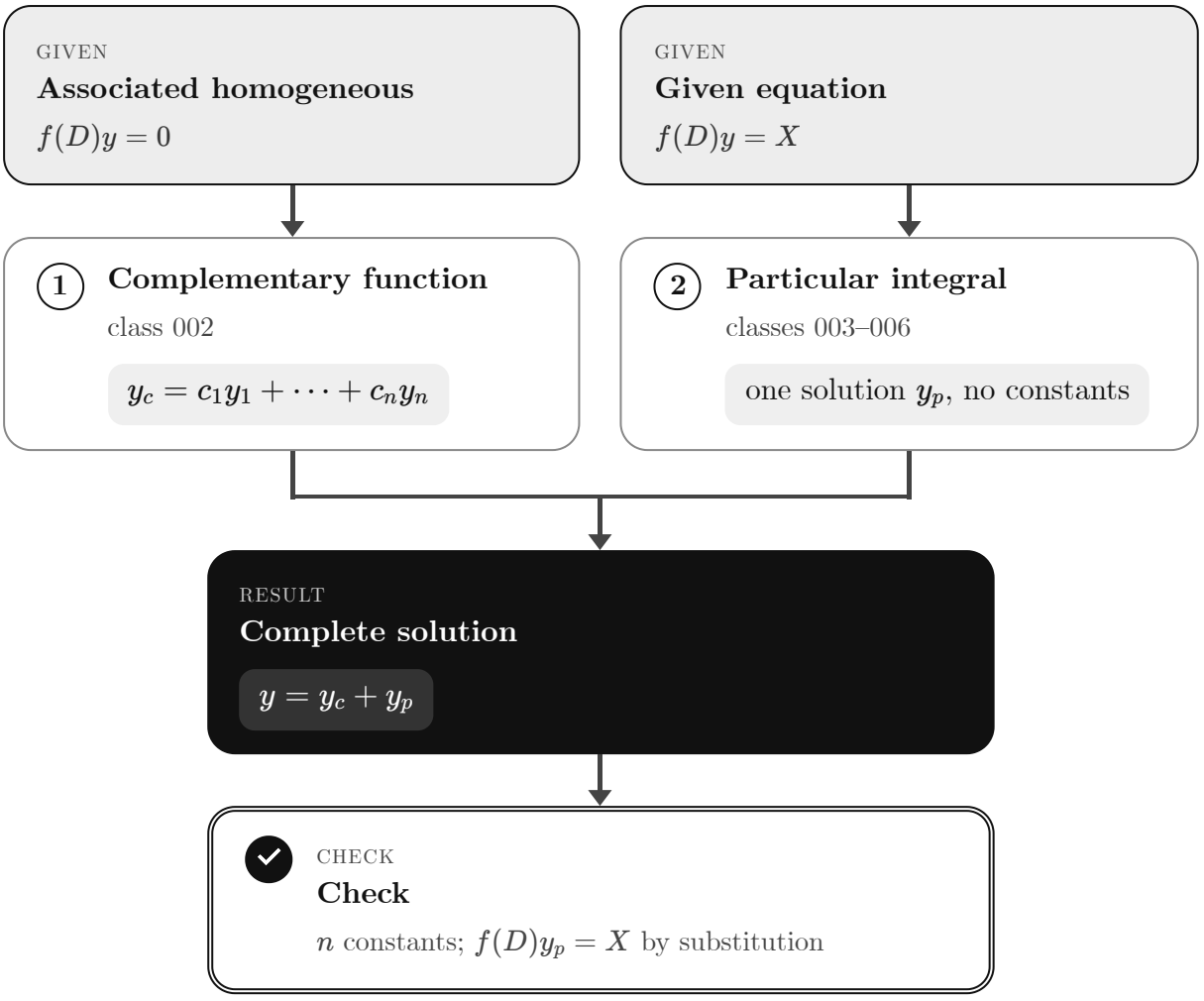


Remark. The Wronskian is added for clarity (not in the deck). For a set of n functions use the definition E4.1 or the $n \times n$ Wronskian.

Figure F4.1. Are two functions linearly independent? ESA ★★

ADDED FOR CLARITY S001·13 · S001·14 · T001 00:22:00

5 The general solution: $y = \text{CF} + \text{PI}$



Remark. Initial conditions, if given, are applied to the complete solution $y_c + y_p$, never to the CF alone.

Figure F5.1. The plan of every Unit 2 answer ESA ★★★

S001·12 · S001·17 · T001 00:18:29 · ESA Dec-23 3a

Complementary function against particular integral

	Complementary function y_c	**Particular integral y_p**
Solves	$f(D)y = 0$ (associated homogeneous)	$f(D)y = X$ (given equation)
Arbitrary constants	n (order of the DE)	none
Form	$c_1y_1 + \cdots + c_ny_n$, y_i independent	one function of x
Unique?	the complete family	no: any two differ by part of the CF
Depends on	$f(D)$ only	$f(D)$ and X
Found in	class 002 (auxiliary equation)	classes 003–006, 011 (variation of parameters)

S001·12 · S001·16 · S001·17

Every Unit 2 ESA question is CF + PI

Kind of question	**Distinct (occurrences)**	**Coefficients**	**Answer form**
PI by operator rules	7 (9)	constant, $f(D)y = X$	CF + PI
Legendre / Cauchy	6 (8)	variable; reduced to constant	CF + PI in z, then back to x
Variation of parameters	4 (5)	constant	CF + PI (PI from W)
LCR circuits	2 (2)	constant, in t	charge = CF (transient) + PI (steady state)
Total	19 (24)		all CF + PI

ESA Dec-23 3a · ESA Jul-23 3c · ESA Jan-24 s3 3b · ESA Jan-24 s3 3d · ESA Dec-23 3c